

Se nos pide verificar la descomposición en valores singulares (SVD) de la matriz A:

$$V^T \cdot V = I$$

$$A = \begin{pmatrix} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{pmatrix}$$

$$A = U \Sigma V^T$$

Verificar las siguientes SVD (completa y reducida) que se proponen.

$$A \cdot A^T = A^T \cdot A = I, \quad U \in \mathbb{R}^{m \times m}; \quad V \in \mathbb{R}^{n \times n}; \quad \Sigma \in \mathbb{R}^{m \times n}$$

$$U: A \cdot A^T \quad \text{y} \quad V: A^T \cdot A$$

$$\bullet A^T \cdot A = \begin{pmatrix} 0 & 4 & 4 & 0 \\ 3 & 1 & 1 & 3 \\ 3 & -1 & -1 & 3 \end{pmatrix} \cdot \begin{pmatrix} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{pmatrix} = \begin{pmatrix} 32 & 8 & -8 \\ 8 & 20 & 16 \\ -8 & 16 & 20 \end{pmatrix}$$

$$\bullet \det(A^T A - \lambda I) = 0$$

$$\begin{vmatrix} 32-\lambda & 8 & -8 \\ 8 & 20-\lambda & 16 \\ -8 & 16 & 20-\lambda \end{vmatrix} = -\lambda \cdot (\lambda - 36)^2$$

$\lambda_1 = 36 \Rightarrow \sigma_1 = \sqrt{36} = 6$   
 $\lambda_2 = 36 \Rightarrow \sigma_2 = \sqrt{36} = 6$   
 $\lambda_3 = 0 \quad (*)$

$$\underline{\lambda = 36} : (A^T \cdot A - 36 I) \cdot \vec{r} = 0$$

$$\begin{pmatrix} 32-36 & 8 & -8 \\ 8 & 20-36 & 16 \\ -8 & 16 & 20-36 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$-4x + 8y - 8z = 0 \Rightarrow x = 2y - 2z$$

$$x - 2y + 2z = 0$$

$$(x, y, z) = (2y - 2z; y; z) = (2y; y; 0) + (-2z; 0; z)$$

$$\{(2; 1; 0); (-2; 0; 1)\}$$

$$V_1 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \quad V_2 = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} \quad \left. \vphantom{\begin{matrix} V_1 \\ V_2 \end{matrix}} \right\} \text{ Normalizada:}$$

$$\|V_1\| = \sqrt{4+4+1} = 3 \quad \|V_2\| = 3 \quad \left\{ \begin{array}{l} V_1 = \frac{1}{3} \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \quad ; \quad V_2 = \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} \end{array} \right.$$

$$V = \frac{1}{3} \begin{pmatrix} 2 & 2 \\ 2 & -1 \\ 1 & -2 \end{pmatrix} \Rightarrow V^T = \frac{1}{3} \begin{pmatrix} 2 & 2 & 1 \\ 2 & -1 & -2 \end{pmatrix}$$

$$\bullet A \cdot \sqrt{\sigma_i} = \sigma_i \cdot \mu_i \Rightarrow \mu_i = \frac{1}{\sigma_i} \cdot A \cdot \sqrt{\sigma_i}$$

$\mu_1 :$

$$\bullet \mu_1 = \frac{1}{6} \cdot \begin{pmatrix} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{pmatrix} \cdot \frac{1}{3} \cdot \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} = \frac{1}{2} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \left. \vphantom{\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}} \right\} U = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$\bullet \mu_2 = \frac{1}{6} \cdot \begin{pmatrix} 0 & 3 & 3 \\ 4 & 1 & -1 \\ 4 & 1 & -1 \\ 0 & 3 & 3 \end{pmatrix} \cdot \frac{1}{3} \cdot \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix}$$

$$A = U \cdot \Sigma \cdot V^T$$

$$A = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 2 & 2 & 1 \\ 2 & -1 & -2 \end{pmatrix}$$

Una posible expresión es:

$$A = \underbrace{\frac{1}{2} \begin{pmatrix} 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \\ 1 & -1 & -1 & -1 \end{pmatrix}}_U \underbrace{\begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{\Sigma} \underbrace{\frac{1}{3} \begin{pmatrix} 2 & 2 & 1 \\ 2 & -1 & -2 \\ -1 & 2 & -2 \end{pmatrix}}_{V^T}$$

$$A^T x = 0$$

$$(1, 0, 0, -1)$$

$$(0, 1, -1, 0)$$

Encuentre la SVD y la pseudo-inversa de las siguientes matrices.

Matriz A

$$\underline{\hspace{2cm}} \bullet \underline{\hspace{2cm}} \quad A = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\bullet A^T \cdot A = \begin{pmatrix} 1 \\ 3 \end{pmatrix} (1 \ 3) = \begin{pmatrix} 10 \end{pmatrix}, \quad \lambda_1 = 10 \Rightarrow \sigma_1 = \sqrt{10}$$

$$\Sigma = \begin{pmatrix} \sqrt{10} \\ 0 \end{pmatrix}$$

$$\bullet \left. \begin{aligned} (10 - 10I) \cdot x &= 0 \\ 0 \cdot x &= 0 \end{aligned} \right\} V = \begin{pmatrix} 1 \end{pmatrix}$$

$$\bullet U: \quad \mu_1 = \frac{1}{\sigma_1} \cdot 4 \cdot \sigma_1 = \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix} \cdot 1 = \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$\mu_2 = \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} -3 \\ 1 \end{pmatrix} \text{ para q' sea ortogonal a } \mu_1.$$

$$U = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & 3 \\ 3 & -1 \end{pmatrix}$$

$$A = U \Sigma \cdot V^T = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} \sqrt{10} \\ 0 \end{pmatrix} \cdot (1)$$

$$\bullet \text{PSEUDO-INVERSA } (A^+) \text{ con } A \in \mathbb{R}^{m \times n}$$

$$A^+ = V \cdot \Sigma^+ \cdot U^T = (1) \cdot \begin{pmatrix} \frac{1}{\sqrt{10}} & 0 \end{pmatrix} \cdot \frac{1}{\sqrt{10}} \cdot \begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix}$$

$$A^+ \cdot A = 1$$

$$\bullet C = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix} \quad C \in \mathbb{R}^{2 \times 3}; \quad C \cdot C^T (2 \times 2) \quad \left\| \begin{array}{l} \text{Hacer } C^T \cdot C \\ (3 \times 3) \end{array} \right\|$$

$$\bullet C \cdot C^T = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix}$$

$$\bullet |CC^T - \lambda I| = 0 \Rightarrow \det \begin{pmatrix} 17-\lambda & 8 \\ 8 & 17-\lambda \end{pmatrix} = (17-\lambda)^2 - 64 = 0$$

$$17-\lambda = \pm \sqrt{64}$$

$$\left. \begin{array}{ll} \lambda_1 = 25 & , \quad \lambda_2 = 9 \\ \downarrow & \downarrow \\ \sigma_1 = \sqrt{25} = 5 & \quad \sigma_2 = \sqrt{9} = 3 \end{array} \right\} \Sigma = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$$

$$\lambda_1 = 25:$$

$$(CC^T - 25I) \cdot \mu = 0 \Rightarrow \begin{pmatrix} -8 & 8 \\ 8 & -8 \end{pmatrix} \cdot \mu = 0$$

$$\mu_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 9 : (CC^T - 9I) \cdot u = 0 \Rightarrow \begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix} u = 0$$

$$u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad U = \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

• Calculamos  $V$ :  $v_i = \frac{1}{\sigma_i} \cdot C^T \cdot u_i$   $V \in \mathbb{R}^{3 \times 3}$

$$v_1 = \frac{1}{5} \cdot \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$v_2 = \frac{1}{3} \cdot \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{3\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}$$

Necesitamos un  $v_3$  que sea ortogonal a  $v_1$  y  $v_2 \Rightarrow$  Prod. Vectorial.

$$v_3 = v_1 \times v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \frac{1}{3\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 4 \end{pmatrix}$$

$$\begin{vmatrix} i & j & k \\ 1 & 1 & 0 \\ 1 & -1 & 4 \end{vmatrix} = 4i - 4j - 2k = (4; -4; -2) \Rightarrow v_3 = \frac{1}{6} \cdot 2 \begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix}$$

$$\|v_3\| = \sqrt{4+4+1} = 3 \checkmark$$

$$V = \begin{pmatrix} 1/\sqrt{2} & 1/3\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -1/3\sqrt{2} & -2/3 \\ 0 & 4/3\sqrt{2} & -1/3 \end{pmatrix}$$

• Pseudo-inversa:  $C^+$

$$C^+ = V \cdot \Sigma^+ \cdot U^T$$

$$C^+ = \frac{1}{45} \begin{pmatrix} 7 & 2 \\ 2 & 7 \\ 10 & -10 \end{pmatrix}$$

$$\Sigma^+ = \begin{pmatrix} 1/5 & 0 \\ 0 & 1/3 \\ 0 & 0 \end{pmatrix}$$

$$U^T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

